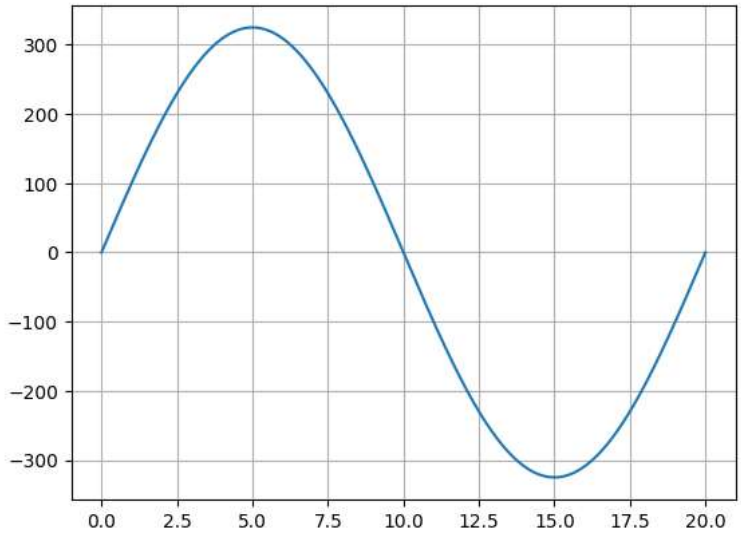
Problem: Using Python and Matplotlib, plot a 50 Hz AC voltage waveform and clearly identify the following quantities on the graph:

1. The instantaneous voltage u(t)
2. The RMS voltage, Ueff=230V
3. The peak voltage, Umax=325V
4. Label the x-axis as time (t) in milliseconds.
5. Label the y-axis as voltage (u(t)) in volts.
6. Give the graph an appropriate title.
7. Add a legend identifying the instantaneous voltage, RMS voltage, and peak value.
8. Add gridlines to make the graph easier to read. The instantaneous AC voltage is represented by u(t)=325*sin(2pi.50.t) where t is measured in seconds. Since the program uses milliseconds,

t_seconds=t_milliseconds *times 10^-3.

```
import numpy as np
import matplotlib.pyplot as plt

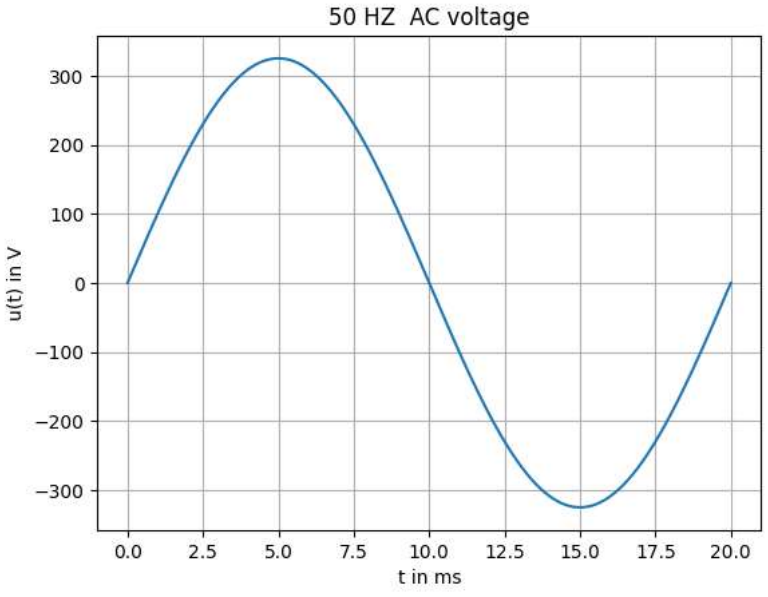
t = np.arange(0, 20, 0.001)       # time in ms
u = 325*np.sin(2*np.pi*50*t*1e-3)
fig, ax = plt.subplots()
ax.plot(t,u)
ax.grid()
plt.show()
```




```
import numpy as np
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t = np.arange(0, 20, 0.001)       # time in ms
u = 325*np.sin(2*np.pi*50*t*1e-3)
fig, ax = plt.subplots()
ax.plot(t,u)
# include xlabel,ylabel and title of the graph
```

```
ax.set(xlabel='t in ms',ylabel='u(t) in V',title='50 HZ  AC voltage')
ax.grid()
plt.show()
```




```
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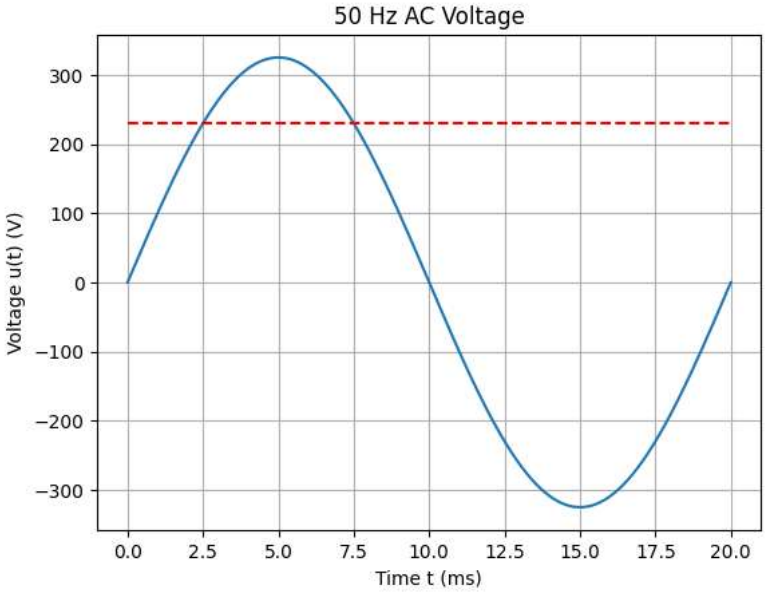
t = np.arange(0, 20, 0.001)       # time in ms
u = 325*np.sin(2*np.pi*50*t*1e-3)
Ueff=[230,230]

fig, ax = plt.subplots()

ax.plot(t, u, label='Instantaneous voltage u(t)')
ax.plot([0,20],Ueff,'r--',label='RMS value:230V')
ax.plot(5,325,'ro',label='peak value:325V')
#ax.axhline(230, linestyle='--', label='RMS voltage = 230 V')
#ax.axhline(-230, linestyle='--')

ax.set(
    xlabel='Time t (ms)',
    ylabel='Voltage u(t) (V)',
    title='50 Hz AC Voltage'
)

#ax.legend()
#ax.legend()
#ax.legend(loc='upper right')
#ax.legend(loc='lower left')
#ax.legend(loc-'best')
ax.grid()
plt.show()
```




```
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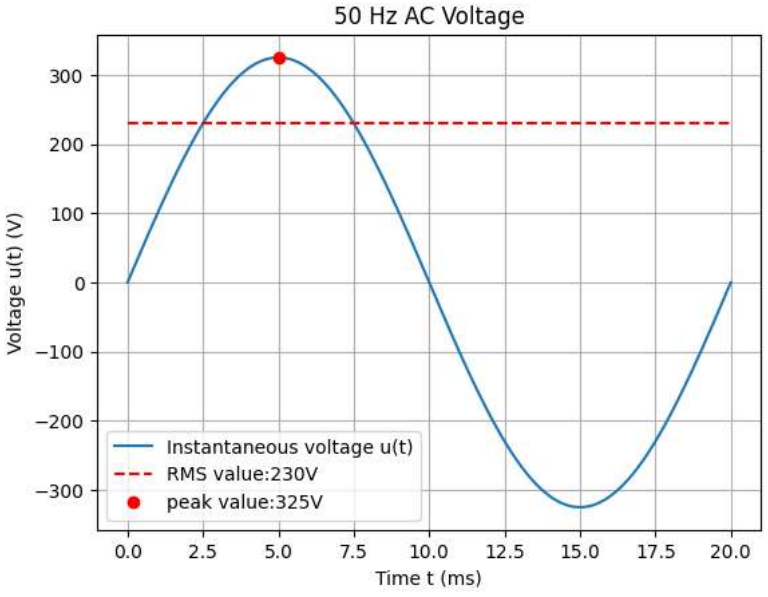
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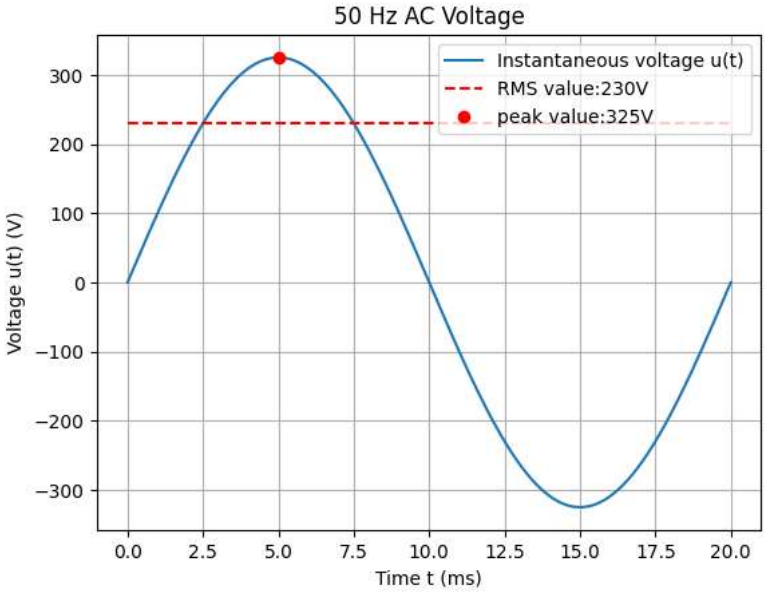
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ax.set(
    xlabel='Time t (ms)',
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    title='50 Hz AC Voltage'
)

#ax.legend()
ax.legend(loc='upper right') # label of the object appear at uppe right corner
#ax.legend(loc='lower left')
#ax.legend(loc-'best')
ax.grid()
plt.show()
```




```
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import matplotlib.pyplot as plt

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u = 325*np.sin(2*np.pi*50*t*1e-3)
Ueff=[230,230]

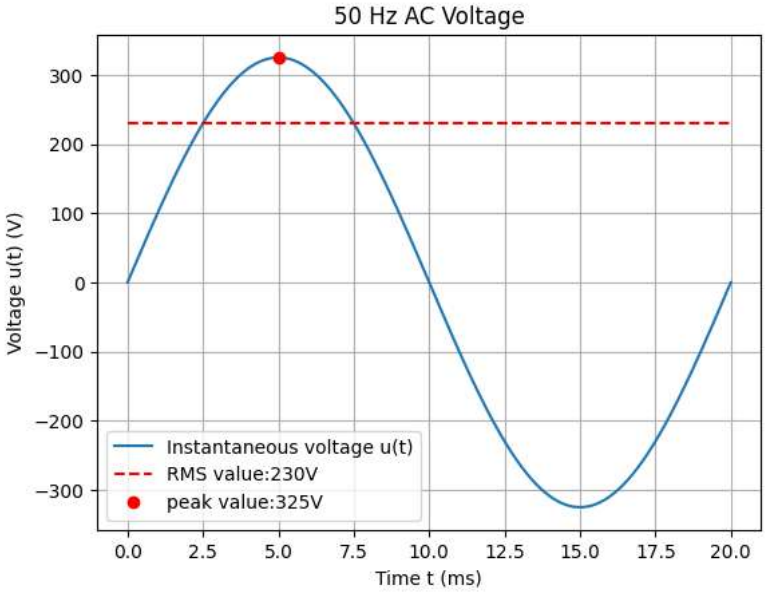
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ax.set(
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    ylabel='Voltage u(t) (V)',
    title='50 Hz AC Voltage'
)

#ax.legend()

#ax.legend(loc='upper right')
ax.legend(loc='lower left') # object label appear at lower left corner
#ax.legend(loc-'best')
ax.grid()
plt.show()
```


50 Hz AC Voltage
Voltage u(t) (V)
Time t (ms)
Instantaneous voltage u(t)
RMS value:230V
peak value:325V
300
200
100
0
−100
−200
−300
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0